

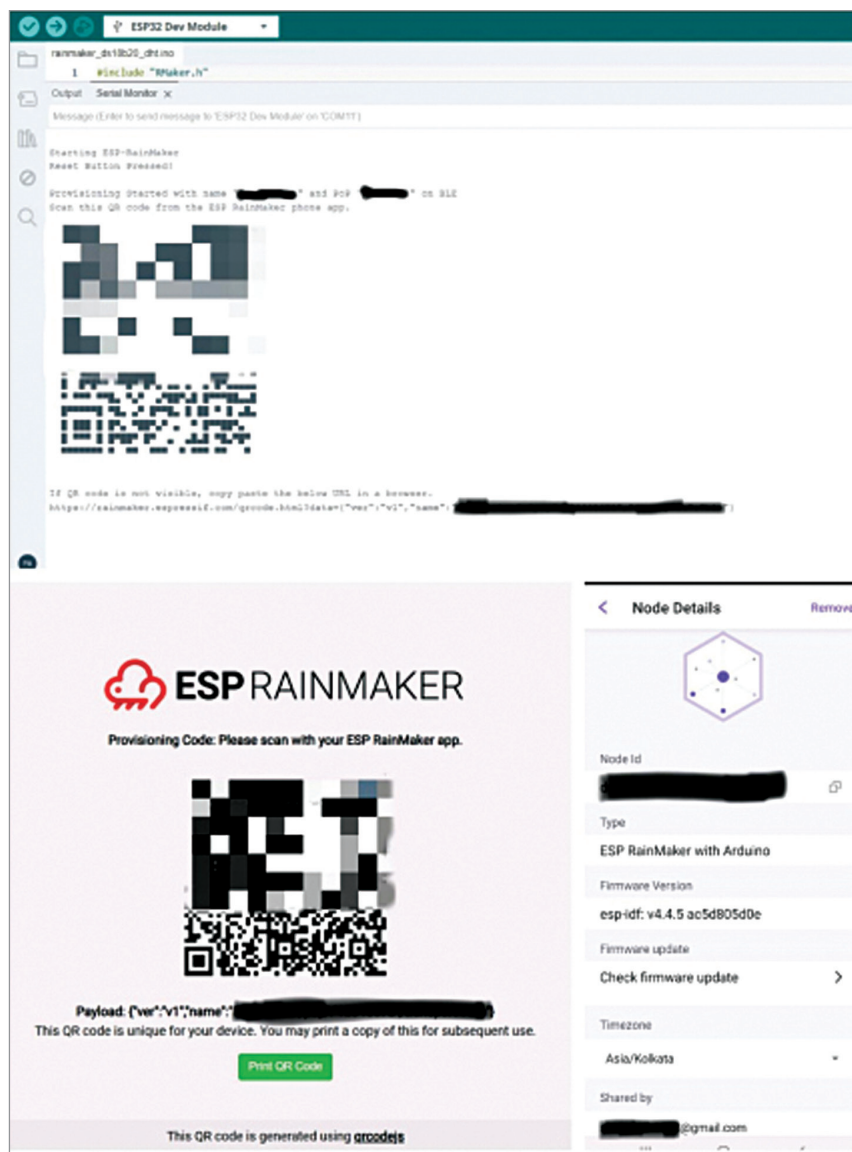


Fig. 5: ESP Rainmaker node setup

firmware upgrade service, which can have parameters such as URL and status.

For the proposed setup, the provisioning and QR code scanning processes, along with node establishment, are illustrated in Fig. 5. If the QR code is not clearly visible, the associated URL can be copied and pasted into a browser for optimal results. The node ID is crucial and must be kept confidential to ensure the correct functioning of the setup. The collection of singular nodes, access

to these nodes, and their operation can be easily shared with other group members through various methods. In this case, this information was shared via email. Fig. 5 depicts the ESP RainMaker node setup.

In the experimental setup, the primary objective was to activate the charging of the Android smartphone

serving as the tethering hotspot. Observations indicated that after approximately 20-21 hours, the smartphone's battery would significantly deplete, requiring the activation of charging. To address this limitation, the on/off hotspot charging button was controlled through the RainMaker app, ensuring that the smartphone maintained a consistent battery level.

For broader applications, a node can be shared among multiple users. In this case, three users, each corresponding to a node, installed the app. This redundancy ensured that if any one node failed to receive an update, the other two nodes could still do so. The current state of the relay (on/off) connected to the charger was reflected on all three smartphones. Additionally, the proposed setup provided updates to all three shared users regarding the relay's on/off status, even if only one of the users operated it via the RainMaker app.

Before uploading the source code, configure the Wi-Fi SSID and password within the code. Then, upload the source code to the ESP32 development board using Arduino IDE, ensuring the correct board and port are selected. Once the code has been uploaded, establish the relay connections as shown in Fig. 3 and detailed in Table 2. Next, download the ESP RainMaker Android app on an Android smartphone. Open the serial monitor to receive the QR code. Scan this QR code with the ESP RainMaker app to connect it to the intended device. Once connected, the app can be used to control the device's charging. **EFY**

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